# Report on "The Large $N$ Limit of Superconformal Field Theories and Supergravity" (J. Maldacena, hep-th/9711200)

## 1. Principal achievement

The paper takes the decoupling limit of $N$ coincident branes (D3, M2, M5, D1–D5, and others) in two ways at once. On the brane side, it gives a superconformal field theory. On the gravity side, it gives the near-horizon geometry $AdS_{p+2}\times$(compact space). The author notes that the curvature in Planck units scales as a negative power of $N$, so for large $N$ supergravity can be trusted even though the brane theory is decoupled. This leads to a bold conjecture. Type IIB string theory on $(AdS_5 \times S^5)_N$ is *exactly dual* to $\mathcal{N} = 4$ $U(N)$ super-Yang–Mills. There are analogous dualities for $AdS_7 \times S^4$, $AdS_4 \times S^7$ and $AdS_3 \times S^3 \times M^4$.

Several consistency checks support the proposal:

- The superconformal group matches the isometry supergroup of the background.
- $SL(2,\mathbb{Z})$ of IIB matches S-duality of SYM, with $g^2_{YM} \sim g$.
- The dictionary $R^4/\alpha'^2 = 4\pi gN$ makes this a strong/weak duality. Supergravity is reliable exactly where the 't Hooft coupling is large and perturbation theory fails.
- The probe D3-brane Born–Infeld action is shown to follow from broken conformal invariance.
- Earlier absorption cross-section and entropy results are reinterpreted.

The paper therefore turns a long-standing hope, a string dual of large-$N$ gauge theory, into a concrete and testable statement. It also gives a non-perturbative definition of quantum gravity in AdS. The radial coordinate $U$ is identified with an energy scale, which is the seed of holographic RG. The impact of this result is hard to overstate.

## 2. Principal drawback

The central claim is a *conjecture* resting on heuristic arguments. The main gap is that the precise map between the two sides is missing. The author explicitly leaves the AdS boundary conditions unspecified ("possibly also some boundary degrees of freedom"), and gives no prescription for how gauge-theory observables (correlators of local operators) correspond to bulk quantities. Without such a dictionary, the conjecture can be tested only through symmetry matching and a few protected quantities, and symmetries alone largely fix those. The paper itself concedes that the probe action "is a consequence of the symmetries and thus not a prediction". The genuinely dynamical tests are either left unchecked (e.g. the existence of strings in the 't Hooft limit) or remain unexplained, such as the 3/4 entropy mismatch and the $N$-dependence for M2 branes. The lower-supersymmetry cases (Sections 4–5, and Reissner–Nordström) are more speculative and carry acknowledged puzzles.

## 3. Most promising follow-up research

The idea that balances feasibility and impact best is to **construct the explicit observable dictionary and use it to make quantitative, non-protected checks**. It has three steps:

1. Identify each single-trace chiral primary operator of $\mathcal{N} = 4$ SYM with a Kaluza–Klein mode on $S^5$, matching dimensions with masses (e.g. $\Delta(\Delta - 4) = m^2R^2$).

2. Propose that the CFT generating functional equals the bulk on-shell action with prescribed boundary values $\phi|_{\partial AdS} = \phi_0$ (with the boundary at large $U$). This also settles the boundary conditions the paper leaves open.

3. Compute two- and three-point functions, and Wilson loops via strings ending on the boundary, at strong coupling.

The work is feasible because it uses only classical supergravity on a known background and the field content and representation theory are already tabulated. It is high-impact because it turns the conjecture into a computational tool. It also provides falsifiable comparisons with protected correlators at weak coupling, and it opens the way to thermal physics (the black hole in Eq. (2.5) as the deconfined plasma) and to non-conformal and confining generalisations.